

Breaking Future Rhythm Visualizer

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ABSTRACT

In breaking, the timing of the technique and rhythm needs to be matched, but the audience has trouble judging whether or not the timing is correct. Therefore, We developed a system that shows the future rhythm of the sound source the DJ is playing.

CCS CONCEPTS

• Applied computing → Sound and music computing.

KEYWORDS

Breaking, beat tracking, sound visualization

ACM Reference Format:

Masatoshi Hamanaka. 2024. Breaking Future Rhythm Visualizer. In *The 37th Annual ACM Symposium on User Interface Software and Technology (UIST Adjunct '24)*, October 13–16, 2024, Pittsburgh, PA, USA. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3672539.3686739>

1 INTRODUCTION

Breaking, the official competition of the Paris Olympics, features two dancers breakdancing against each other to the rhythm of music played by a DJ (disc jockey). The DJ uses two turntables to play multiple musical pieces while switching between them. The DJ then adds accents to the music by scratching the turntable. Freezing while performing tricks such as inverted stands in perfect time with the rhythm generated by the DJ is called "Kill the beat" and is the highlight of the competition, as it requires a high level of skill. Most of the audience at the venue of breaking have experienced breaking, and they get very excited when they see a successful kill the beat. However, viewers who see breaking for the first time have trouble judging the success of kill the beat on TV or video. Therefore, we have developed a Breaking Future Rhythm Visualizer, which displays the future rhythm of the sound source being played by the DJ (Fig. 1).

Several methods were proposed to automatically generate a setlist for DJs to prepare before remixing[1, 7, 10]. An automated DJ-mixing system was also proposed[8] as well as a method for smoothly crossfading from one piece to another[11].

While these studies[1, 7, 8, 10, 11] intended to automate the DJ's work or assist the DJ, our study is intended to assist the breaking audience through visualization. Rhythm visualization helps the audience understand if the rhythm matches the timing of the moves.

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UIST Adjunct '24, October 13–16, 2024, Pittsburgh, PA, USA

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ACM ISBN 979-8-4007-0718-6/24/10

<https://doi.org/10.1145/3672539.3686739>



Figure 1: Breaking Future Rhythm Visualizer

Visualization will also enable the hearing impaired to understand the timing of the rhythm and enjoy breaking.

Our Breaking Future Rhythm Visualizer was used on February 2024 at the All Japan Breaking Championships, which were recorded by the Japan Broadcasting Corporation (NHK in Japanese). The recorded video can be viewed on the NHK's website.

2 BREAKING FUTURE RHYTHM VISUALIZER

There are three problems in visualizing the future rhythms of music generated by a DJ. Each of these problems and their solutions are herein described.

Future rhythms are difficult to predict: Predicting future rhythms is difficult because DJs change musical pieces one after another. Therefore, we receive all sound sources used by DJs, analyze them, and generate rhythm visualization videos. Then, the sound source and the video can be played synchronously by creating a link between the sound source and the video. When a DJ changes the music playback speed (turntable rotation speed), the video playback speed also changes in sync.

Appropriate musically The rhythm visualization video to be created consists of long bars for strong beats and short bars for weak beats, and it scrolls vertically with time (Fig. 1). The strong and weak beats can be obtained by AI analysis. However, simple analysis by the AI sometimes results in the strong and weak beats being reversed, depending on the timbre of the instrument or other factors. Therefore, we added a musical analysis to solve the problem of strong and weak beats being reversed [2–6].

Video and signal delays: The broadcast video produced by NHK includes delays as it passes through selectors that integrate multiple cameras and switchers that superimpose information, etc. The distance between the cameras and the broadcast room at NHK Hall, the venue of the convention, was more than 100 meters. In addition, the distance between the cameras and the broadcast room at the

NHK Hall, the venue of the Championships, was more than 100 meters, which causes transmission delays. However, our Breaking Future Rhythm Visualizer also needs to send information from the DJ booth to the broadcast room, which causes delays. Therefore, due to these delays, the timing between the dance video and the rhythm visualization video is shifted. Therefore, we can adjust the position of the rhythm visualization video by adding a line indicating the current time on the video later in the program so that the timing of the dance video and the rhythm visualization video are aligned.

Animation at current timing: Adding an effect such as a color change or animation when the strong or weak beat bars pass the current timing makes the timing easier for the viewer to understand. However, a video with effects added in advance cannot be created because the timing and position change depending on the delay in the previous item. Therefore, we developed a program that analyzes the rhythm visualization video being played, that reads the bar positions, and that adds effects at the appropriate time.

Detection of turntable rotation Two turntables are connected to the DJ mixer, and DJ software can be used to set the pieces assigned to each turntable. When using DJ software, a vinyl record, in which sine waves with different phases are recorded on the left and right sides, is placed on the turntable. The DJ software receives the rotation information of the turntable from the digital vinyl system (DVS) signal generated by the vinyl record. The DJ software then receives the turntable rotation information from the DVS signal generated.

Because the DVS signal is an analog signal, we branch it out, receive it at the audio interface, and analyze it to determine the RPM of both the left and right turntables. Specifically, accelerating the turntable increases the frequency, and decelerating the turntable decreases the frequency; because the DVS signal is a stereo signal and is out of phase between left and right, it can also detect whether the rotation is forward or reverse depending on which waveform arrives first.

Scratch detection DJ mixers have a crossfader, which is used for a swap between two pieces or for adding scratching sounds. Scratching is a technique in which a turntable is operated to rub the other turntable during the playback of a certain piece to emphasize the rhythm with the sound generated. The crossfader can then be moved slightly to the turntable on the side that is being scratched to mix both the playback and scratch sounds. Therefore, to detect scratches, we need to detect not only the rotation of the turntable but also the position of the crossfader.

Many models of DJ mixers that can be connected to a computer via a USB cable use MIDI signals for communication. In addition, some DJ mixer models that are equipped with two USB ports for DJ alternation can receive the USB signal flowing through the port in use on the other USB port. Therefore, the MIDI signal is analyzed, and the crossfader position is extracted.

3 IMPLEMENTATION

Figure 2 shows an overview of the system. We aimed to achieve the desired function by combining general-purpose equipment without using special equipment. Turntables rotation information is obtained by branching the audio cable that connects the turntable

to the mixer and by inputting it to the audio interface connected by a universal serial bus (USB) cable to a Mac mini. The USB cable is connected between the mixer and the Mac mini to acquire the crossfader position. The Mac mini then detects the scratch from the rotation and crossfader information, which is then transmitted to the Mac studio via MIDI signals. However, the video of the rhythm

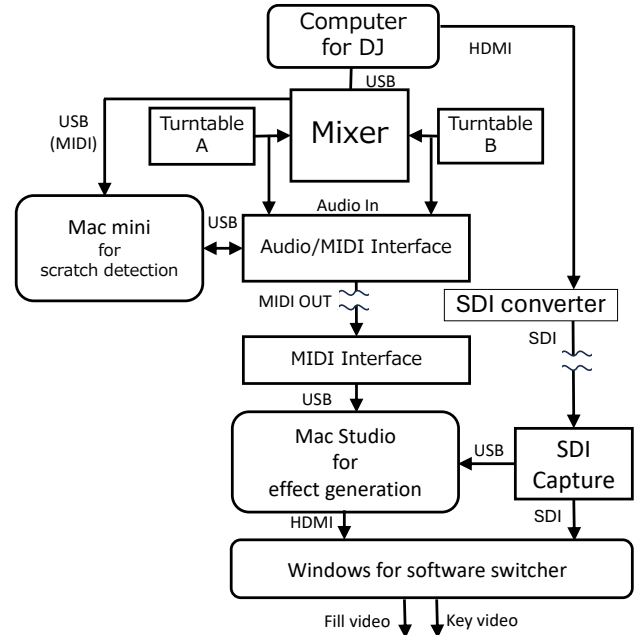


Figure 2: Overview of the Future Rhythm Visualizer.

visualization output from the computer used by a DJ is captured by the Mac studio, and image analysis is performed to detect when the bar passes the pixel set as current. The Mac studio then draws the scratch and the animation of the bar as it passes. Finally, the video drawn by the Mac studio and the rhythm visualization video are captured by the Windows machine to generate the fill video and key video for the video composition.

4 CONCLUSION

The Breaking Future Rhythm Visualizer we constructed was used at the Japan Breaking Championships in February 2024. The video was recorded using a sound source and its rhythm visualization video after the video was superimposed. Slow playback of the recorded video showed a discrepancy between the timing of the sound in the broadcast video and the timing of the rhythm visualization. Therefore, the current point of the visualization video was adjusted to eliminate the misalignment. The recorded video can be viewed on NHK's website [9], (in Japanese). When twenty people who viewed the recorded video were asked their opinions, 18 said they enjoyed watching it, and 14 said "Kill the beat" was easy to understand. We plan to conduct more detailed experiments.

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